

# Why Java?

- Problems in c/c++

portability(platform dependent) -> we have to use **compiler** multiple times for the same code as every platform gave different binary code (010101001011) , so there was no portability.

**compiler** = Turns code into machine code (binary) as machine can't understand Normal code.

- Problem with compiler

->it used to translate the code for only **one platform**, So only one platform will understand , therefore we have to compile the same code

again and again for different platforms which was not portable.

Hence c/c++ is **not a portable** language.

**Now what is a platform?**

Platform consists of two things i.e. processor and OS

combination of processor and OS is termed as Platform

**Java is a platform Independent Language**

which means It doesn't have to be compiled again and again.

**BYTECODE**

**JVM(Java virtual Machine) "Translator"**

JVM stands for Java Virtual Machine. It is a virtual machine that enables a computer to run Java programs as well as programs written in

other languages (such as Kotlin, Scala, or Groovy) that are compiled into Java bytecode.

we can use Bytecode on any Platform as JVM is already installed on Platform.

JVM is platform Dependent

BYTECODE is platform independent

JAVA is platform independent

## JAVA IS PORTABLE

platform independent is also termed as **WORA (Write once , Run Anywhere)**.

## ==SECURITY

- Backend-> Servlet
- Frontend-> Applets(discontinued)
- Client-> Web Browser

Also,JVM provides a Secured Environment , this is called a Sandbox Model.

**JVM SOLVES 2 PROBLEMS -> PORTABILITY AND SECURITY**